

Department of Electrical Engineering & Electronics

Year 1 PhD Project Plan Form

Student Name	Ignacio Martin Allati	Student ID	201292757
Primary Supervisor Name	Dr. Xin Tu	Secondary Supervisor Name	Prof. James Bradley

Project Plan

Introduction

Polycyclic Aromatic Hydrocarbons (PAHs) have been reported to be universally ubiquitous representing close to 15% of the elemental carbon in the Interstellar Medium (ISM). These molecules might have an impact in the heating and ionization balance of the interstellar gas. Moreover, PAHs might be involved in stars formation. While there is strong evidence for their presence in space, their numerous roles in chemistry and physics of the ISM are yet to be fully understood.

One of the biggest questions underlying this research issue relates to the reaction pathways of PAHs in space. What species could be derived from them? What other reactive species could be involved in the decomposition of PAHs? What are the reaction mechanisms of this decomposition?

To the best of our knowledge, most research efforts have been aimed towards the role of PAHs on earth as pollutants affecting atmosphere and aquatic environments, and how to catalytically decompose them. However, not many publications can be found regarding the key role of PAHs in the ISM. Hence, as an attempt to address this challenge, a study of the reaction pathways of PAHs under plasma condition will be conducted including both experimental and theoretical approaches. Indeed, plasma processing is a key feature of this research project as it enables to recreate the state of the matter and the chemical environment of the ISM.

Overall, the ultimate objective of this whole PhD project is to reveal the reaction mechanisms of PAH decomposition in the ISM.

Plan for Year 1

One of the pillars of the first year will be Plasma Kinetic Modelling. This theoretical approach contributes to the understanding of reaction mechanisms once a realistic model of the reactive system has been built. To the best of our knowledge, such plasma kinetic model of PAHs decomposition has not yet been attempted as no publications can be found on this matter.

In this first year, a Zero Dimensional model of a Dielectric Barrier Discharge reactor (0D-DBD) will be developed for this project and implemented by means of ZDPlasKin Fortran code. This model will be first applied to simpler systems like CO₂ splitting, Dry Reforming, Steam Reforming, among others. Not only these reactions are relevant for the energy and environmental fields as very active research topics, but also this first step will test the model and allow the candidate to gain experience on it. The model's results will be matched against readily available experimental data of these reactions performed on DBD reactors.

The development of a model for PAHs will be kept as a long term objective because of its complexity, i.e. not all the information is available to represent the chemical behaviour of PAHs. For instance, reaction constants will be obtained in collaboration with partners of the EUROPAH Project working on ab initio calculations. Also, experimental data of PAH decomposition to test the model is still to be collected, either as part of this PhD project or in collaboration with EUROPAH project colleagues.

The second pillar of this year consists of experimental works on low pressure plasma processing of PAHs and its chemical precursors (potentially Acetylene). Low pressure plasma will be generated by radio frequency sources. Dust of various catalysts will be included in the system, interestingly with a composition similar to that of the interstellar dust. In this way, these experiments will be highly relevant to the chemistry of the ISM.

Whenever possible in terms of workload, DFT calculations will be included in this first year for a deeper and complementary understanding of the plasma kinetic results. Details of these studies are still to be defined (the systems to be studied and the code to be employed).

Objectives

The objectives and work plan to be achieved in the first year are listed as follows:

1st September 2017 – 22nd December 2017

- Literature review on plasma kinetic modelling.
- Learn about plasma systems, in particular cold atmospheric plasma and DBD reactors
- Learn how to use ZDPlasKin Fortran Code and gain experience with it.
- Develop user code, set up a logic of input and output files useful for data analysis.
- Build plasma kinetic model for CO₂ Splitting and perform calculations to match available experimental data.
- Implement code in HPC facilities
- Attend a number of mandatory seminars and lectures as part of both PhD program and EUROPAH Project.

2nd January 2018 – 16th April 2018

- Write first publication on CO₂ Splitting – Plasma kinetic modelling.
- Build plasma kinetic model for Dry Reforming and perform calculations to match available experimental data.
- Develop a housekeeping subroutine to reduce reaction sets as the calculation progresses. Try to exploit parallelism in the code to run it in several cores simultaneously.
- Literature review for experimental works to be performed subsequently.
- Training for experimental works to be performed subsequently.
- Perform experiments on RF activated low pressure plasma processing of PAHs and or PAHs precursors, with or without catalyst dust.
- Literature review on DFT applied to plasma systems.
- Attend a number of mandatory seminars and lectures as part of both PhD program and EUROPAH Project.

19th April 2018 – 30th August 2018

- Write second publication on Dry reforming – Plasma kinetic modelling.
- Build plasma kinetic model for Steam Reforming and perform calculations to match available experimental data.
- Further literature review on PAHs' chemistry.
- First approach to develop a plasma kinetic model of PAHs (based on previous progress such as housekeeping, parallelism, literature review and colleague's experimental results).
- Perform DFT calculations for a deeper/complementary insight on the systems modelled so far.
- Attend a number of mandatory seminars and lectures as part of both PhD program and EUROPAH Project.

Project Supervision

Please, describe how the supervision is being and will be conducted. For example, frequency of meetings, timing of feedback, communication with the supervisory team, any other expectations, etc.

Progress will be monitored by means of monthly formal meetings held with the primary project supervisor, Dr Xin Tu. A formal record of these meetings shall be kept through the PGR toolbox. In addition, meetings with the secondary supervisor Prof James Bradley shall be held at least every 6 months for additional guidance and advice.

Various informal meetings with both Dr Xin Tu and Prof James Bradley may take place on a weekly basis to discuss progress and seek for advice.

Group meetings may also take place, providing an opportunity to discuss topics relevant to the research and receive feedback from colleagues.

An annual report of the first year's progress shall be produced at the end of the academic year.

Facilities and Resources

All facilities and resources are available to complete the project.



Yes



No

If no, please explain how these will be obtained.

Induction

The student has attended a Departmental PhD Induction event.



Yes



No

Please state the date of the event:

08/09/2017

Risk Assessment

The student has completed the Departmental Safety Risk Assessment.



Yes



No

Please state the date of the assessment:

11/09/2017

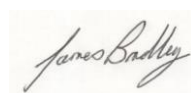
**Primary Supervisor
Signature**



Date

14/12/2017

**Secondary Supervisor
Signature**



Date

14/12/2017

**Student
Signature**



Date

14/12/2017

Additional Comments

The Progress Assessment Panel (PAP) members for the student will provide feedback on this plan.